

Notes for Hand Tapping Workshop

November 2025

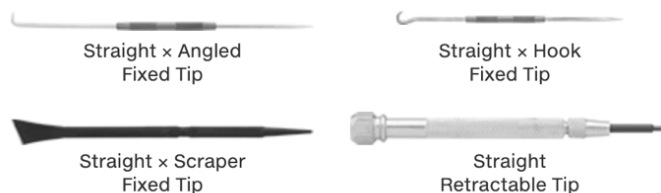
Before you TAP you will need to accurately mark your material. This is sometimes referred to as “layout”. You will also need something for making measurements (a ruler or tape measure) and a **COMBINATION SQUARE**. Note: you can use “layout dye” when transferring marks onto aluminum or metal (I sometimes use a BLUE SHARPIE instead of layout dye).

To do this, you will need a **SCRIBE** and a **CENTRE PUNCH** or PUNCH.

There are different **SCRIBES** for different materials. I typically use one scribe as a multi-purpose tool. See this link for examples of different types of SCRIBES:

<https://www.mcmaster.com/products/layout-scribers/scribers-1~/>

Scribers for Steel and Plastic



Use the steel tip on these scribers to mark steel and plastics, or as a pick for cleaning and probing. They have a knurled grip for a secure hold.

Retractable Tip—Retractable tip stays inside the handle for protection when it's not in use.

Once you have marked where you want to drill your hole, you can use a **CENTRE PUNCH** to place a dimple in your material. This will help to accurately position your DRILL BIT. See this link for examples:

<https://www.mcmaster.com/products/center-punches/center-punches-3~/>

Round Handle



A typical CENTRE PUNCH

There are different types of DRILL BITS for different material. Here are some examples:



Forstner bit (wood)



Hole Saws
Connect to a drill or drill press to make circular cuts

32 products



Hole Saw Arbors
Secure a hole saw to a drill or drill press

15 products

Hole saws (aluminum, brass, bronze, steel, wood)



Spade (wood)



Brad Point (wood)



Masonry

High-Speed Steel Drill Bits

Cut holes in a variety of metals, from the softest aluminum to the hardest steel



High-Speed Steel (HSS) bits are **SUPER COMMON**

High-Speed Steel Drill Bits for Plastic



Penetrate gradually to prevent cracking and chipping

Similar to HSS bits but with more of a tapered point.

Drilling your hole.

We will use a HSS (high-speed steel) drill bit to drill a hole in your acrylic material.

Always CLAMP your work and place a piece of wood under your acrylic to prevent BREAKOUT/CHIPPING. Use cutting fluid when you drill into acrylic. If you don't have fancy cutting fluid, a bit of water will do the trick.

Choose a drill bit that matches the size of the TAP you want to use. **Here is a guide:**

Tap Size	Drill Size (number)	Drill Size (fractional)
4-40	43	5/64
6-32	36	7/64
8-32	29	9/64
10-24	25	
10-32	21	11/64
12-24	16	
¼-20	7	7/32

Use a drill press to drill your hole. Always clamp your work! Using a drill press ensures that your hole is straight (90 degrees to the surface of your material).



Use a drill press to drill a straight hole in your material.

Once you have drilled your hole, you are now ready to TAP!

There are three common taps: **Taper, Plug and Bottoming**:

Bottoming Tap (1 - 1.5 Tapered Threads)



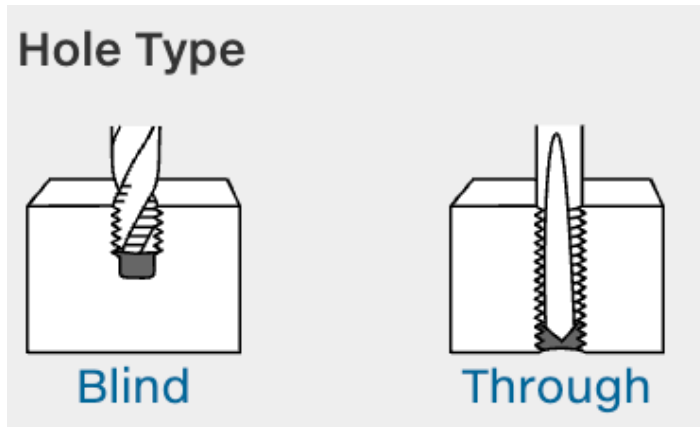
Plug Tap (3 - 5 Tapered Threads)



Taper Tap (8 - 10 Tapered Threads)



What tap you choose depends on the type of hole you have drilled in your material:



Use a **TAPER** or **PLUG** Tap to start threading a “BLIND” hole and then finish the job with the **BOTTOMING** Tap. This will ensure that your threads make it to the bottom of your hole.

Use a **TAPER** Tap for threading a hole that runs **THROUGH** the material.

You will want to use a GUIDE to help you start your thread and make sure it is entering the hole nice and STRAIGHT. You can buy a Tap Guide or make your own out of a block of wood or block of aluminum or acrylic.



Tap Guides help you to keep your hand tool straight while you start your tap.

You will want to cut your threads **slowly and gradually**. You are pushing/turning the tool into the material. Use **cutting fluid** if you are tapping metal or aluminum.

TURN the TAP a couple of turns CLOCKWISE and then REVERSE a couple of turns. This process of going forward and then reversing helps to clear CHIPS from the HOLE. This helps to prevent the Tap getting stuck in the HOLE. If this happens, your Tap may break! You want to avoid this!!

Some Taps are designed so that they SELF CLEAR the debris buildup from the hole. These Taps do not need to be reversed. These Taps are called **SPIRAL POINT TAPS**.



Spiral-Point Flute for Through Holes

Here is a nice video that explains Spiral-Point Taps:

<https://www.youtube.com/watch?v=mwKloqgkgyo>

Combination Square: A great stocking stuffer for the holidays!

Combination Squares

Replace your try square, miter square, depth gauge, marking gauge, 90° level, ruler, and scribe with one tool. Blades labeled with numeric increments that are less than 1" are quick-reading.

Blades with Square Head with Built-In Level

